

Practice Sheet – 1

Probability and Statistics

BMAS 0108

Ques. 1 What is the average miles per gallon (mpg) for all new cars? A random variable sample of 35 new cars gave an average of 21.1 mpg.

- a) Identify the variable
- b) Is the variable quantitative and qualitative?

Ques 2. Below the list of variables (and their possible values). Which of these variables are quantitative and which are qualitative? Out of the four possible scale of data (Nominal, ordinal, interval, ratio), which scale does each of them belongs to? Explain your answer.

- a) The rating of a movie.
- b) Birthday
- c) Weight
- d) Height.
- e) The total debt of the Italian government in 2010.
- f) The stock exchange on which a company stocks traded.
- g) Your birth place
- h) The letter grades in your first-year courses.
- i) Amount owing in your credit card.
- j) The amount you spend per month entertainment.
- k) Amount of rainfall in a month.
- l) The number of courses you are taking.

Ques 3. The following data give the lifetime of 30 incandescent light bulbs (rounded to the nearest hour) of a particular type.

872	931	1146	1079	915	879	863	1112	979
1120	1150	987	958	1149	1057	1082	1053	1048
1118	1088	868	996	1102	1130	1002	990	1052
1116	1119	1028						

Construct frequency, relative frequency and cumulative relative frequency table.

Ques 4. A biologist assumes that there is a linear relationship between the amount of fertilizer supplied to tomato plants and the subsequent yield of tomatoes obtained.

Eight tomato plants, of the same variety, were selected at random and treated, weekly, with a solution in which x grams of fertilizer was discovered in a fixed quantity of water. The yield, y kilograms, of tomatoes was recorded.

plant	A	B	C	D	E	F	G	H
X	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
Y	3.9	4.4	5.8	6.6	7.0	7.1	7.3	7.7

a) plot a scatter diagram of yield, y , against amount of fertilizer, x .

Ques 5. Construct a histogram, frequency polygon, and ogive using relative frequencies for the distribution (shown here) of the miles that 20 randomly selected runners ran during a given week.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	6
15.5–20.5	3
20.5–25.5	9
25.5–30.5	4
30.5–35.5	7
35.5–40.5	5

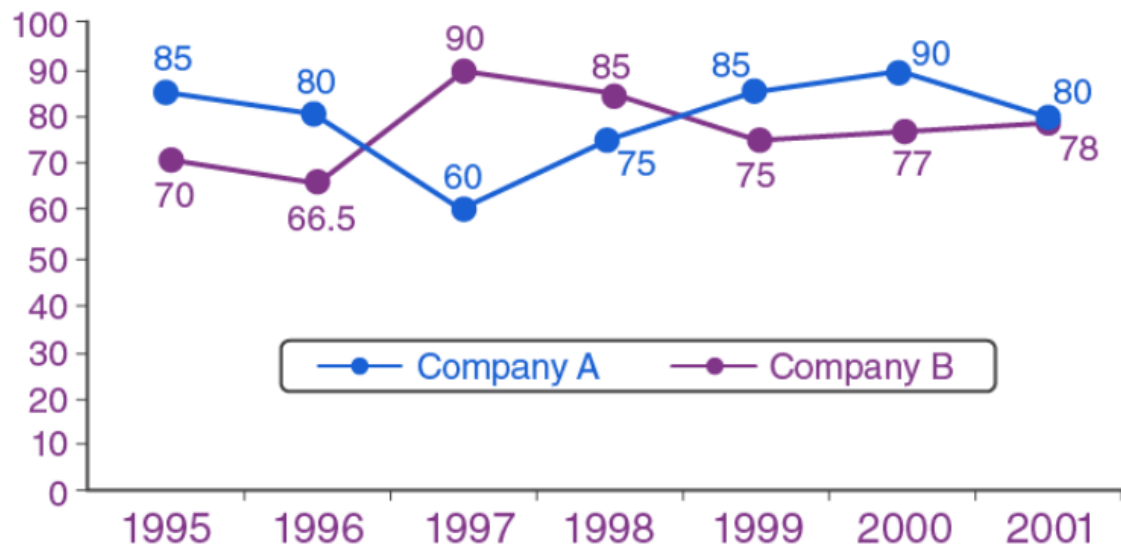
Ques 6. The data show the number of railroads crossing accidents for the 50 states of the United States for a specific year. Construct a histogram, frequency polygon, and ogive for the data.

Class limits	Frequency
1–43	24
44–86	12
87–129	3
130–172	14
173–215	1
216–258	2
259–301	4
302–344	3

Ques 7. The mean and variance of eight observations are 9 and 9.25 respectively. If six of the observations are 6, 7, 10, 12, 12, and 13, find the remaining two observations.

Ques 8. The following graph represents the profit percentage earned by 2 companies A and B on their investment. [Hint: Revenue = Investment + Profit]

Line graph questions – 2



Go through the line graph and answer the following questions.

- In which year, the ratio of investment and profit is maximum for company A?
- In which year, the investment is minimum for company B?

Ques 9. The table represents Akshay's weight in kilograms for five months. Draw the line graph for the information provided.

Month	Akshay's weight in kilogram
January	48
February	53
March	60
April	68
May	72